Example of Combinations with Repetition

- Suppose that a cookie shop has 4 different kinds of cookies. In how many different ways can 6 cookies be chosen? There are no constraints on how many cookies of each kind are chosen (e.g., all 6 may be of the same kind)
- **Solution**
  Equivalently, I wish to distribute 6 indistinguishable cookies (stars) to 4 distinguishable cookie boxes (bars).
- By previous theorem, the number of such combinations is
  $$C(4+6-1,6) = C(9,6) = C(9,3) = 84$$
- Is the solution still correct if some cookie box contains fewer than 6 cookies? (Equivalently, the cookie shop does not have 6 cookies for some cookie kind(s)).
- No. We cannot place 6 stars in that box. Solution later …
- Equivalent problem: Find the number of (nonnegative integer) solutions for equation: $x_1 + x_2 + x_2 + x_4 = 6$, where $0 \le x_i \le 6$ indicates the chosen number of cookies of type *i*.
- The number of solutions is $C(4+6-1,6) = C(9,6) = C(9,3) = 84$